Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

Proposer	**Name**	Henning Heuer	**Email**	Henning.heuer@ikts.fraunhofer.de
Institute Name	Fraunhofer IKTS Dresden			
Address	Maria-Reiche Str. 2 01109 Dresden Germany			

2. Technology Overview

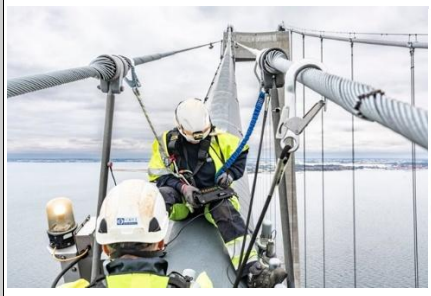
Title of Proposed Technology	**Imaging Ultrasonic Drones / Hoppers for Testing Critical Infrastructure**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (x), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		Electrical and Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Safety Monitoring/Diagnostic Instrumentation and Control Systems	Measuring Instruments	Measurement Sensors and Components

Concept of Proposed Technology

Non-destructive testing (NDT) using drone-mounted systems represents a groundbreaking approach for inspecting large and difficult-to-access critical infrastructures such as bridges, storage tanks, wind turbines, and other industrial installations. By integrating unmanned aerial vehicles (UAVs) with ultrasonic sensors and other NDT modalities, this technology enables remote material inspection without the need for direct human access via climbing, aerial work platforms, or scaffolding. Currently crawler-based systems for autonomous climbing / moving are in industrial use, but with its system immanent limitation of accessibility just to top side or obstacle free areas.

Left: State of the art inspection crawler for full area ultrasonic imaging scans from Fraunhofer IKTS for top side inspection e.g. wind turbine blades [1]; Right: Ultrasonic drone with a single element ultrasonic probe, for point-by-point wall thickness measurements

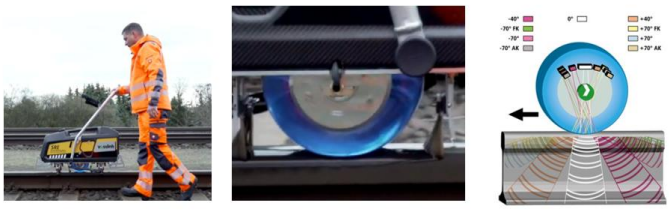
Current Drone-based NDT systems are limited to simple single probe solutions that allows a point-by-point in flight approach but no full area scans or muti angle imaging that is fundamentally required for NDT based determination of structures integrity. This lack of practical applicability, sufficient NDT performance, and reliable data quality inhibits drone-based NDT from technical and economic success until now. One key reason is the adaptation of existing conventional NDT components, such

	as ultrasonic, eddy current, or thermal systems, to existing drone platforms. To overcome these limitations, Drone NDT systems must be fundamentally reengineered and fully optimized for aerial operation. This includes the development of lightweight, drone-compatible multichannel imaging sensors, practicable and robust positioning and stabilization systems, and data acquisition and analysis frameworks tailored for dynamic inspection environments. **Strategy** The proposed idea is a combination of crawler technology with its benefits of scanning Ultrasonic (Phased Array) imaging and modern quadcopter flexibility. In many countries and regions, drone operations are subject to very strict regulatory constraints. To enable broad, low-threshold applicability, an **ultrasonic hopping device (hopper) or hopping crawler** concept is proposed.  **Present and future of ultrasonic NDT for critical and large infrastructure** In this approach, the testing system overcomes obstacles through short-range, low-height hops that are not classified as flight operations. The inspection itself is performed via rolling wheel contact, while the required contact force is provided by propellers. This approach also enables suspended or overhead (inverted) operation. Wheel based crawlers allow a vibration free stable sensor approach and movement (scanning) over a structure for full area imaging whereas quadcopters allow flexible access to difficult object geometries.
Rationale for Joint R&D Project	The proposed project requires strong interdisciplinary collaboration and faces several challenges that have so far hindered purely privately funded development. At the same time, the potential impact and post-project benefits are highly relevant for society. Within the project, Fraunhofer IKTS will be responsible for the development of the non-destructive testing (NDT) technology. Potential partners specializing in autonomously mountable, separable manipulators such as crawlers and climbers are required, along with an end-user and NDT service provider (applicant).
Key Words	1. Safety of Critical Infrastructure 2. Smart Robots and Drones 3. Non-destructive Testing

3. Research and Development Objectives & Plan

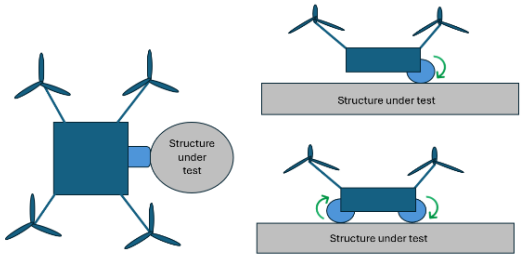
Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (5)	**End of the Project**	Level (8)
R&D Objectives	**New Idea and development task** **1. Phased Array or Multiprobe Ultrasonic Probe Wheel** Building on Fraunhofer IKTS competences in the field of mobile ultrasonic rail track inspection, we propose an ultrasonic probe wheel containing flexible configurable phased array or multichannel probes. An NDT wheel offers two key advantages over conventional contact techniques: (1) compliance to irregular surfaces via a soft rubber tire that maintains coupling, and (2) the ability to perform line			

scans with minimal water usage and minimal wear of the probe face.



Ultrasonic probe wheel developed by Fraunhofer IKTS for rail track inspection [2].

The existing ultrasonic testing (UT) probe wheel must be adapted for drone applications, regarding weight, size and probe configuration. The wheel can be partially housed within the drone body, where coupling gel is dispensed onto the wheel surface for acoustic coupling. The UT wheel itself is used as guidance for stable drone movement on the structure under test.



Principle concept of drone with Ultrasonic probe wheel for Phased Array or multiple angle probes,
Top: front wheel position better accessibility to complex structures),
bottom: wheel pair for more stability and contact pressure control

2. Lightweight Multichannel Ultrasonic Electronics

Modern NDT applications require B-scan or C-scan imaging, typically produced with phased-array or multi-angle configurations, to detect and size defects in accordance with current regulations and state-of-the-art practice. Consequently, multiprobe or phased-array solutions are essential. To perform a true 3-D volume scan, or achieve full angular coverage, 10 to 64 phase-synchronous channels are usually needed. Current NDT-drone prototypes, however, are still limited to single-channel methods.
Fraunhofer IKTS has its own ultrasonic system platform that has been deployed successfully in industrial environments for more than 20 years, with over a thousand units in service [3]. The 64 and 128 channel system weight 2 to 4 kg are sensitive to vibrations and have a solid metallic case for EMV shielding and heat dissipation.

The Fraunhofer PCUS© Family (left) and the 128 full parallel system PCUS Pro A2

For airborne applications, a 64-channel unit, at one-quarter the size of current systems and weighing about 500 g, including shielding and a polymer-based heat sink, must be developed and fully validated for NDT use under flight conditions. In parallel, energy consumption must be optimized. A battery-powered 10-channel system paired with a tablet currently delivers about **3 h of operation**, but it requires a **3 kg battery pack**. For a complete drone-borne system, we must first define the allowable payload, the expected productive flight time, and the required NDT performance, and then size the power supply accordingly.

R&D Target Specifications	The final project objective is the realization of an operable prototype of an imaging ultrasonic NDT hopper, suitable for remotely controlled operation on outdoor industrial structures. The system will be equipped with a multichannel ultrasonic probe wheel operating in the 2–5 MHz frequency range, covering an angular spectrum from −30° to +30°, with a spatial resolution and sensitivity of approximately 1 mm. The internal delay line of the probe wheel is adjustable for testing metallic structures with thicknesses ranging from 5 to 100 mm. The software platform enables motion control, ultrasonic system parameterization, and three-dimensional data visualization.

R&D Plan

Task	Partner	26Q3	26Q4	27Q1	27Q2	27Q3	27Q4	28Q1	28Q2
NDT Concept	all								
Electronics	IKTS								
Probe Wheel	IKTS								
Control SW	IKTS								
Hopper Concept	Partner								
Fight module	Partner								
Drive Module	Partner								
Merging	all								
Field Application	all								

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

WP 1: Development of a Phased-Array / Multiprobe Ultrasonic Inspection Wheel

- Design and development of an integrated phased-array or multiprobe ultrasonic wheel for continuous NDT
- Selection, arrangement, and mechanical integration of ultrasonic transducers
- Optimization of acoustic coupling concepts
- Evaluation of resolution, penetration depth, and scanning speed
- Prototype implementation and experimental validation under realistic conditions

WP 2: Battery and Power Supply System for the NDT Module

- Development of an energy-efficient battery and power supply concept for mobile NDT applications
- Selection of suitable battery technologies considering runtime, weight, and safety
- Integration of charging, protection, and monitoring electronics
- Power consumption analysis of all subsystems
- Implementation of a modular and scalable power architecture

WP 3: Control System for NDT Wheel Parameterization

- Development of a control system for configuration and monitoring of inspection parameters
- Parameterization of ultrasonic frequency, transmit power, focus, scan geometry, and sampling rate
- Implementation of closed-loop control to adapt to material and surface variations
- Interface development between hardware, sensors, and software
- Validation of measurement stability and reproducibility

WP 4: Lightweight Multichannel Ultrasonic Electronics

- Design of compact, lightweight multichannel ultrasonic electronics
- Implementation of parallel transmit and receive channels for phased-array operation
- Integration of position sensing (e.g. encoders, IMU) for accurate data localization
- Implementation of intelligent power management to reduce energy consumption
- EMC-compliant design and thermal optimization

WP 5: NDT Software for Control, Data Acquisition, and Parameterization

- Development of modular software architecture for system control
- Implementation of configuration and parameterization functions for ultrasonic inspection
- Real-time data acquisition and signal preprocessing (e.g. filtering, gain, TOF)
- Visualization of inspection results (A-, B-, and C-scans)

Expected R&R of Korean SME Partner

- Develop concepts for applying drone / hopper technology to industrial facilities.
- Design and manufacture a scanner or crawler equipped with non-destructive equipment.
- Develop remote control and data management programs for integrated systems.
- Develop ultrasonic probes (if necessary).
- Develop application-specific technical applications and calibration procedures.
- Preliminary performance verification and safety evaluation of the non-destructive testing system using ground-based or alternative platforms

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The ultimate outcome of this international joint R&D project is to demonstrate that drone-based non-destructive testing (NDT) can advance to a technological level suitable for practical industrial applications. Through this project, Fraunhofer IKTS aims to overcome the structural and technical limitations inherent in existing drone-based NDT approaches and to establish a new technological architecture that meets the data quality and reliability requirements for the inspection of industrial facilities and civil infrastructure. This project addresses the current limitations of drone-based NDT systems, which are largely restricted to single-sensor, point-by-point measurements, making them unsuitable for comprehensive structural integrity assessment. By enabling area-based scanning (line-scan approach) and multi-angle, multi-channel ultrasonic imaging (B-scan / C-scan), the project seeks to fundamentally expand the technological applicability of drone-based NDT systems. **1. Expected Technological Outcomes** 1) Experimental validation of drone-based area-scanning ultrasonic NDT concepts Through this project, Fraunhofer IKTS will validate an integrated technological concept addressing the most critical challenges in combining a flight platform (drone) with contact-based ultrasonic testing, namely: ● Stable maintenance of ultrasonic coupling ● Controlled contact pressure during inspection ● Precise sensor movement and attitude stability during operation To this end, Fraunhofer IKTS will adapt its existing wheel-based ultrasonic probe technology for drone applications by reducing weight and size and optimizing the structure to maintain inspection quality on curved, rough, coated, or corroded surfaces. This achievement represents a key technological transition, elevating drone-based NDT from an auxiliary measurement tool to a practically applicable inspection solution. 2) Verification of the applicability of lightweight multi-channel ultrasonic electronics for drone platforms For reliable structural integrity assessment, multi-channel or phased-array ultrasonic imaging is essential, rather than single-channel ultrasonic measurements. Based on its existing multi-channel ultrasonic system platforms, Fraunhofer IKTS will define the concept of lightweight, drone-compatible multi-channel ultrasonic electronics and verify their applicability under flight conditions. The project will deliver the following technological outcomes: ● Definition of acceptable constraints for weight, power consumption, thermal management, and electromagnetic shielding in drone environments ● Identification of the limitations of ultrasonic data quality under flight-induced vibration, attitude variation, and power constraints ● Establishment of design guidelines for multi-channel ultrasonic systems applicable to aerial platforms These results constitute core technological assets that can be extended beyond drones to a wide range of mobile and robotic NDT systems. 3) Establishment of expandability toward collaborative drone-based ultrasonic inspection concepts While the project focuses primarily on single-drone contact-based inspection in the short term, it clearly establishes the potential for future expansion toward the collaborative drone concept (separated transmitter and receiver) proposed by Fraunhofer IKTS. This approach enables access to inspection domains previously unattainable by conventional drone-based NDT, such as: ● Evaluation of internal structural conditions ● Assessment of embedded reinforcement or tendon conditions within concrete structures As such, the project provides a strategic pathway for extending Fraunhofer IKTS's existing collaborative robotic ultrasonic inspection platforms to aerial inspection systems. 4) Core intellectual property and technology transfer–oriented outcomes Through the execution of this project, Fraunhofer IKTS will accumulate key technological assets, including: ● Structural design know-how for drone-adapted, wheel-based ultrasonic probes ● Design and validation criteria for multi-channel ultrasonic systems on aerial platforms

- Methodologies for ultrasonic inspection quality assessment under flight conditions
- Jointly developed intellectual property and documented technological results

In particular, the project serves as a practical validation case for concrete technology transfer scenarios, including joint design activities, prototype verification, and the transfer of testing protocols.

2. Expected Commercial and Industrial Impact

The outcomes of this project will address the current lack of industrial trust in drone-based NDT technologies and contribute to positioning them as viable solutions for real-world inspection markets.
Specifically, the project will enable:

- The transition of drone-based NDT from experimental or auxiliary tools to solutions aligned with industrial inspection standards and requirements
- Improved safety, efficiency, and data quality in the inspection of large-scale and hard-to-access structures
- Strategic expansion of Fraunhofer IKTS's ultrasonic inspection technology portfolio into next-generation aerial and robotic NDT domains

These impacts will play a critical role in maintaining and strengthening Fraunhofer IKTS's technological leadership in the field of advanced non-destructive testing.

3. Long-term Impact

In the mid- to long-term, this project is expected to generate the following broader impacts:

- Verification of the practical commercialization potential of drone-based NDT technologies
- Introduction of a new inspection paradigm through the convergence of aerial platforms, robotics, and ultrasonic testing technologies
- Establishment of a foundation for standardization activities and follow-up large-scale international R&D projects

Ultimately, the project will serve as a strategic stepping stone for expanding Fraunhofer IKTS's non-destructive testing technologies into the aerial inspection market, beyond the scope of a single R&D effort

6. References

[1] https://www.ikts.fraunhofer.de/en/industrial_solutions/ultrasonics/test_systems.html

[2] https://www.vossloh.com/en/products-and-solutions/product-finder/product_43968.php

[3] Elektronik - Fraunhofer IKTS